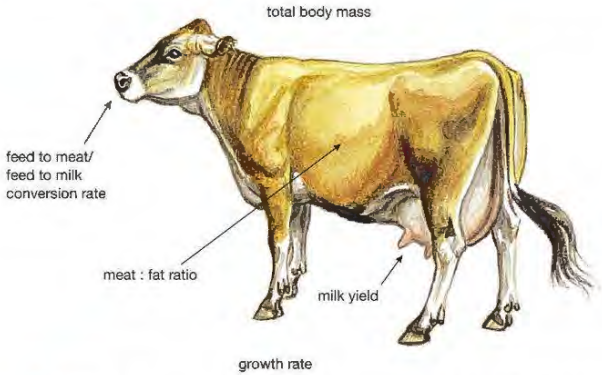
SKILLS ANALYSIS (7)

7 The diagram shows some of the features of a cow that might be used as a basis for a breeding programme.



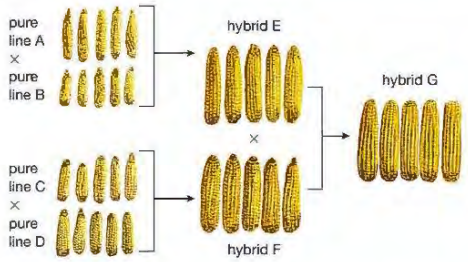
a Which features would you consider important in a breeding programme for dairy cattle?

SKILLS REASONING (8)

b Assume that you had all the techniques of modern selective breeding available to you. Describe how you would set about producing a herd of high-yielding beef cattle.

SKILLS ANALYSIS (9)

8 The diagram shows the results of a breeding programme to improve the yield of maize (sweetcorn).



a Describe the breeding procedure used to produce hybrid G.

(6)

b Describe three differences between the corn cobs of hybrid G and those of C.

SKILLS REASONING (9)

c How could you show that the differences between hybrid G and hybrid C are genetic?

SKILLS INTRAPERSONAL (9)

9 Carry out some research and write an essay about the benefits and concerns of selective breeding of animals. You should write about one side of A4.

UNIT QUESTIONS

SKILLS REASONING (9) **1**

For natural selection to operate, some factor has to exert a 'selection pressure'. In each of the following situations, identify both the selection pressure and the likely result of this selection pressure.

a Near old copper mines, the soil becomes polluted with copper ions that are toxic to most plants. **(2)**

b In the Serengeti of Africa, wildebeest are hunted by lions. **(2)**

c A farmer uses a pesticide to try to eliminate pests of a potato crop. **(2)**

(Total 6 marks)

BIOLOGY ONLY

SKILLS CRITICAL THINKING (7) **2**

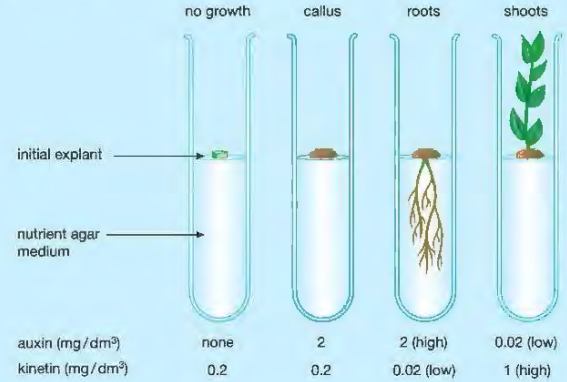
Micropropagation produces thousands of genetically identical plants. Small 'explants' from the parent plant are grown in culture media.

a Outline the main stages in micropropagation. **(4)**

SKILLS REASONING (7)

b Explain why the plants formed by micropropagation are genetically identical. **(2)**

c In some cases, the explants used contain only a few cells, without roots or shoots. Plant hormones are added to the culture media to encourage root and shoot formation. Two of these hormones are called kinetin and auxin. The diagram shows the effects of using different concentrations of the two hormones on root and shoot growth of the explants.



	no growth	callus	roots	shoots
auxin (mg/dm³)	none	2	2 (high)	0.02 (low)
kinetin (mg/dm³)	0.2	0.2	0.02 (low)	1 (high)

(7)

i What is the effect of adding kinetin or auxin without any other hormone? **(2)**

ii Describe how you would treat these explants to produce first shoots and then roots. **(3)**

SKILLS CRITICAL THINKING (7)

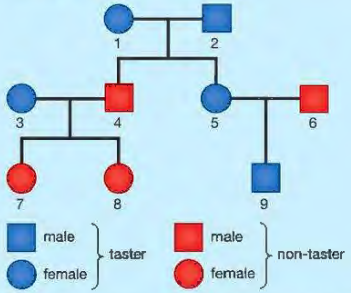
d Explain one advantage and one disadvantage of micropropagation. **(2)**

(Total 13 marks)

END OF BIOLOGY ONLY

SKILLS ANALYSIS 9

3 PTC (phenylthiocarbamide) is a chemical that to some people has a very bitter taste, while other people cannot taste it at all. The diagram shows the inheritance of PTC tasting in a family.



a What evidence in the diagram suggests that the allele for PTC tasting is dominant? **(2)**

SKILLS REASONING 9

b Using T to represent the tasting allele and t to represent the non-tasting allele, give the genotypes of individuals 3 and 7. Explain how you arrived at your answers. **(4)**

SKILLS ANALYSIS 9

c Why can we not be sure of the genotype of individual 5? **(2)**

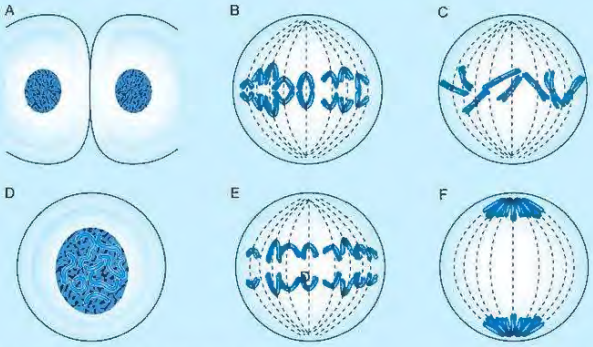
SKILLS INTRPRETATION 10

d If individuals 3 and 4 had another child, what is the chance that the child would be able to taste PTC? Construct a genetic diagram to show how you arrived at your answer. **(4)**

(Total 12 marks)

SKILLS ANALYSIS 7

4 The diagrams A to F show an animal cell during cell division. The diploid number of this cell is eight.



a Put the pictures in the correct order. **(2)**

8

SKILLS CRITIACL THINKING 6

b Is the cell going through mitosis or meiosis? Explain your answer. **(2)**

7

c What is the diploid number of a human cell? **(1)**

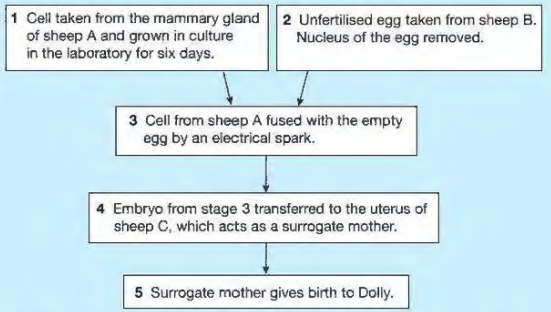
d Describe two differences between mitosis and meiosis. **(2)**

(Total 7 marks)

BIOLOGY ONLY

SKILLS CRITICAL THINKING 8

5 The following flowchart shows how Dolly the sheep was cloned.



a Where did scientists get the DNA to put into the unfertilised egg from sheep B? **(1)**

6

b How does the nucleus removed from an egg differ from the nucleus of an embryo? **(1)**

SKILLS ANALYSIS 8

c Dolly is genetically identical to another sheep in the diagram. Which one? **(1)**

SKILLS CRITICAL THINKING 8

d Give two ways in which this method is different from the normal method of reproduction in sheep. **(2)**

e Suggest two advantages of producing animal clones. **(2)**

(Total 7 marks)

7 **6** Copy and complete the following passage about genes:

A gene is a section of a molecule known as _____ .
The molecule is found within the _____ of a cell, within thread-like structures called _____ .
The strands of the molecule form a double helix joined by paired bases. The base adenine is always paired with its complementary base _____ , and the base cytosine is paired with _____ . During the process of transcription, the order of bases in one strand of the molecule is used to form _____, which carries the code for making proteins out to the cytoplasm.

(Total 6 marks)

SKILLS PROBLEM SOLVING 9

7 In a section of double-stranded DNA there are 100 bases, of which 30 are cytosine (C).

Calculate the total number of each of the following in this section of DNA:

a complementary base pairs

b guanine (G) bases

c thymine (T) bases

d adenine (A) bases

e deoxyribose sugar groups **(Total 5 marks)**

END OF BIOLOGY ONLY